

IT367 – Software Project Management

Lab 7: PERT/CPM and Critical Path Analysis

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Project Title	Abhinay: Film Production Networking and Casting Platform	Date	21-09-2026
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Aim

To prepare a project activity network, calculate ES, EF, LS, LF and slack, and identify the critical path and total project duration.

Very Simple Reminder

- Predecessor = activity that must finish before another activity can start.
- ES (Earliest Start) = earliest time an activity can start.
- EF (Earliest Finish) = ES + duration.
- LS (Latest Start) = latest time an activity can start without delaying the project.
- LF (Latest Finish) = LS + duration.
- Slack = LS – ES (or LF – EF).
- Critical activity = activity with zero slack.
- Critical path = connected path of critical activities. A delay on this path can delay the project.

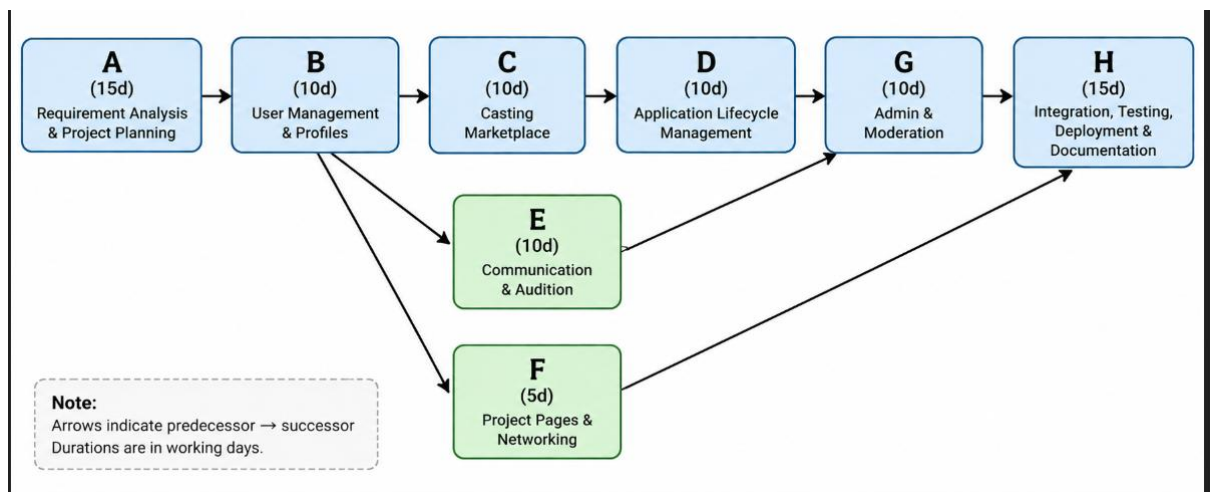
Part A – Input from Your Own Project

Use the SAME project used in earlier labs. Select 8–12 dependent activities from Lab 6. Do not create a new imaginary project.

Activity ID	Activity Name	Duration (days)	Predecessor(s)	Reason / Assumption
A	Requirement Analysis & Project Planning	15	None	Lab 6 Activity 1 = 3 weeks; assume 5 working days/week.
B	User Management & Profiles (1.1)	10	A	Lab 6 Activity 2 = 2 weeks; starts after Activity 1.

C	Casting Marketplace (1.2)	10	B	Lab 6 Activity 3 = 2 weeks; depends on User Management.
D	Application Lifecycle Management (1.3)	10	C	Lab 6 Activity 4 = 2 weeks; depends on Casting Marketplace.
E	Communication & Audition (1.4)	10	B	Lab 6 Activity 5 = 2 weeks; can run after User Management.
F	Project Pages & Networking (1.5)	5	B	Lab 6 Activity 6 = 1 week; can run after User Management.
G	Admin & Moderation (1.6)	10	D, E	Lab 6 Activity 7 = 2 weeks; requires D and E.
H	Integration, Testing, Deployment & Final Documentation	15	F, G	Lab 6 Activity 8 = 3 weeks; requires F and G.

Part B – Draw the Activity-on-Node Network



Part C – Forward Pass and Backward Pass

Complete the table. Show at least one forward-pass calculation and one backward-pass calculation.

Act.	Duration	ES	EF	LS	LF	Slack	Critical?
A	15	0	15	0	15	0	Yes

B	10	15	25	15	25	0	Yes
C	10	25	35	25	35	0	Yes
D	10	35	45	35	45	0	Yes
E	10	25	35	35	45	10	No
F	5	25	30	50	55	25	No
G	10	45	55	45	55	0	Yes
H	15	55	70	55	70	0	Yes

One forward-pass calculation: $ES(B) = EF(A) = 15$ days; $EF(B) = ES(B) + 10 = 25$ days.

One backward-pass calculation: $LF(G) = LS(H) = 55$ days; $LS(G) = LF(G) - 10 = 45$ days.

Part D – Final Result

Critical activities: A, B, C, D, G, H

Critical path: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow G \rightarrow H$

Total planned project duration: 70 working days (14 weeks)

Which activity/activities need the most attention? A, B, C, D, G and H, because they have zero slack and delays can directly delay project completion.

If one critical activity is delayed by 2 days, project completion may be delayed by 2 working days unless time is recovered through corrective action or resequencing.

Part E – 2–3 Sentence Reflection

The PERT/CPM analysis identifies A–B–C–D–G–H as the critical path with zero slack. It helps the project manager prioritize monitoring of these activities while using the available slack in E and F to manage resources and schedule risk.

CPM Assumption: 1 week = 5 working days. Durations are converted from the Week 1–14 schedule in Lab 6; the predecessor relationships are taken from that schedule.